

The National Sustainability Protection Policy

Purpose:

The purpose of this policy is to incentivise companies across Canada to use better methods of manufacturing and design, driving the change towards meeting Canada's carbon emission goals, while also giving Canadians the ability to repair their products with greater ease.

Problem:

Today, many products in Canada are designed with proprietary components such as specialized screws and permanent adhesives that prevent consumer maintenance and third-party repairing. As a result, repairing the product is far too expensive and practically impossible. This issue impacts Canada youth, as they want to make sustainable financial and environmental choices, but these market designs are forcing them to discard broken products, generating an escalation of waste in Canadian landfills. As a result, many Canadian youth of today feel betrayed by their government because of the lack of impactful climate action.

Scope:

When brought into effect, this policy will directly affect all manufacturers and distribution companies operating within Canada, and will indirectly affect all consumers within Canada.

Exceptions:

- a) All Indigenous businesses that would normally act in accordance with this policy are given special permissions to opt-out if they so choose.
- b) Companies with a net income that is less than \$10,000 CAD are exempt from any terms outlined by this policy, but are required to abide when this threshold is crossed at the start of the given quarter.

Main Directives:

1. Impose a per-unit tax on manufacturers that engineer products with proprietary parts such as custom screws and superglue.
2. Direct all the collected tax to the businesses that successfully follow the policy guidelines, using standard repairable parts and recyclable materials to make products both sustainable and accessible.
3. Require companies to give their consumers access to repair. This includes public access to basic tools, simple repair guides, and affordable replacement parts.
4. To make recycling easier on a national level by requiring material labeling and standardized packaging.
5. To extend the products' lifespans so fewer broken goods get thrown out, which reduces waste in landfills.

Definitions:

Company - Any distributor, manufacturer or business of any kind that operates in Canada and is responsible for the creation of new products in any way

Net Income - The total income of a given company in the previous quarter

Policy - Refers to this document itself at all other points mentioned

CRA - Canadian Revenue Agency

ARU - Atomic Recyclability Unit

CSU - Component Standardization Unit

Product - A physical good created by a company for sale to the general public in Canada

Component - An independent part of a product, whose material or function differs enough for it to be individually distinguishable

NCR - National Component Registry

QR - Quick Response, usually in reference to a type of interactive code

Procedures:

a) *Registration:*

Prior to distributing goods in Canada, the manufacturers are forced to submit a detailed schematic and a material breakdown of their engineered product to the Global Component Registry. The governing body then evaluates the product and issues a Recyclability Score alongside a public QR Code to be displayed on the retail packaging

b) *Tax Calculation:*

When a company wishes to create a new product, this policy mandates that they shall send the design plans to the Canadian Revenue Agency for further review. The company must also disclose the intended retail price in Canadian Dollars of the product to the CRA. If the CRA lacks details, the company must comply and provide this information upon request. Failure to comply is considered tax evasion, and shall be treated accordingly.

The CRA should then generate an ARU for the given product like so:

$$P_{ARU} = \sum_{i=1}^n \left[\frac{M_i}{M_p} \times \sum_{j=1}^m \left(\frac{m_{ij}}{M_i} \times \left[\frac{(V_{ij} \times B_{ch}^{\circ}(S_{ij})) + \Delta H_{R,ij}}{MW_{ij}} \right] \times \left(1 + \frac{E_{a,ij}}{E_{ref}} \right) \times T_{ij} \right) \right]$$

Where:

P_{ARU} is the product's ARU

n is the number of components in the product

M_i is the weight of component i

M_p is the weight of the product

m is the number of pure substances in any component

m_{ij} is the weight of pure substance ij

V_{ij} is the virgin material value (0 - 1) for the pure substance ij

B_{ch}° is the standard chemical exergy of S_{ij}

S_{ij} is the pure substance ij

$\Delta H_{R,ij}$ is the enthalpy of fusion for metals, and the enthalpy of decomposition for all else, determined by pure substance ij

MW_{ij} is the molecular weight of the pure substance ij

$E_{a,ij}$ is the activation energy (kJ/mol) of ij

E_{ref} is defined as a reference activation energy of 40 kJ/mol

T_{ij} is the toxicity multiplier of ij

The CSU for a given product should be calculated like so:

$$P_{CSU} = \sum_{i=1}^n \left(\frac{M_i}{M_p} \times \frac{1}{\log_{10}(V_{nat(i)} + 1.5)} \right)$$

Where:

P_{CSU} is the product's Component Standardization Unit

n is the number of NCR-indexed components in the product

M_i is the weight of component i

M_p is the weight of the product

$V_{nat(i)}$ is the number of component i 's occurrences in the NCR

Finally, the total tax per unit should be calculated like so:

$$T_P = P_R \times \left(\frac{P_{ARU}}{ARU_{max}} \right) \times P_{CSU} \times T_{max}$$

Where:

T_P is the total taxation per unit for product P (CAD)

P_R is the product's intended retail price (CAD)

P_{ARU} is the product's Atomic Recyclability Unit

ARU_{max} is the maximum threshold for an ARU score, defined as 20, but subject to change

P_{CSU} is the products Component Standardization Unit

T_{max} is a restraining value set at 50%, but subject to quarterly change under the guidance of the CRA

c) *Manufacturing:*

The company is permitted to manufacture their given product, but must report how many units were manufactured such that the information is given to the CRA at the latest of 2 weeks before the next quarter starts.

d) *Branding Requirements:*

On any given product whose designs are submitted to the CRA, it may be requested that a QR code leading to the National Component Registry homepage be added to the packaging. If so, the company must abide. Failure to do so is treated as a violation of this policy and shall be treated accordingly. This requirement will vary from product to product, as some have no packaging, or in some cases it is infeasible to complete this request.

e) *Taxation:*

At the beginning of each quarter, a company will be taxed by the CRA for each unit of each product manufactured in that quarter, as defined by the product's tax value (defined in section b).

f) *Funding:*

Taxes collected in any given quarter should be used to pay off any civil lawsuits brought against companies, and the remaining value should be distributed amongst companies, such that each company receives a portion of the

tax pool based on the average scores of their products in the previous quarter and their total quarterly income compared to the summation of all quarterly incomes.

Responsibilities:

- a) It is the responsibility of all companies to act in accordance with this policy, and all the terms outlined in the various sections.
- b) It is the responsibility of the CRA to make public the score data collected from all companies on a quarterly basis, unless a given company has applied for a reporting exemption and been approved.
- c) It is the responsibility of the CRA to lead a continuous education campaign that informs companies about how to act in accordance with this policy.
- d) It is the responsibility of the CRA to provide information to companies when requested.

Compliance & Enforcement:

If any company is found to have evaded the taxes outlined in this policy by any means, they shall be held responsible in a civil lawsuit, where the CRA will sue the company for a value amounting to the tax evaded and up to 5% of the company's net income. If any other conflict arises, it is up to the CRA to pursue an investigation and prosecute the perpetrating party.

Reporting Requirements:

The CRA shall be required to report all applicable information about the scores of various companies in an annual report, and to the National Component Registry website. The information reported should include a list of companies on which this policy applies, and information regarding the taxation pool value at the end of each quarter.

Related Legislation:

Bill C-244: A federal Canadian legislation amending copyright laws to allow individuals and third parties to circumvent digital locks for the purpose of maintaining or repairing products.

Canadian Environmental Protection Act, 1999 (CEPA): The main federal law that lets the Canadian government regulate waste, restrict single-use plastics, and enforce strict nationwide pollution standards.

Quebec Bill 29: A provincial legislation that makes it illegal for companies to design products meant to break early. It sets a mandatory multi-year legal warranty (ranging from three to six years) on common electronics and household appliances.